P. P. DIHYDROCODEINE TABLET
Dihydrocodeine tartrate 30mg

COMPOSITION:
Each tablet contains dihydrocodeine tartrate 30mg.

DESCRIPTION:
A white to off white coloured, round shaped tablet.

ACTIONS & PHARMACOLOGY:
Dihydrocodeine is a semisynthetic narcotic analgesic, related to codeine, which acts mainly on the central nervous system and organs with smooth muscle components. The main therapeutic action of dihydrocodeine is analgesia. It has minimal sedative or hypnotic actions. Besides being a potent analgesic, dihydrocodeine has also been used as a cough suppressant. Dihydrocodeine exhibits extensive first-pass metabolism either in the gut wall or liver, and oral bioavailability is only about 20%. In one study, peak plasma concentrations after an oral dose of 30mg was reached at 1.6 hours, and elimination half-life was about 4 hours.

INDICATION:
P. P. Dihydrocodeine is indicated for the relief of painful disorders such as headache, dysmenorrhea, conditions involving musculoskeletal pain, myalgias and neuralgias. It is also indicated as an analgesic and antipyretic in conditions accompanied by discomfort and fever, such as the common cold and viral infections. P. P. Dihydrocodeine is an effective analgesic after dental work and tooth extractions.
Codeine is indicated in patients older than 12 years of age for the treatment of acute moderate pain which is not considered to be relieved by other analgesics such as paracetamol or ibuprofen (alone).

RECOMMENDED DOSAGE:
Dihydrocodeine is best administered with or after food.

Adults:
One tablet (30mg) every 4 to 6 hours.

Paediatric population:
<u>Children aged less than 12 years:</u>
Codeine should not be used in children below the age of 12 years because of the risk of opioid toxicity due to the variable and unpredictable metabolism of codeine to morphine.
P. P. Dihydrocodeine is contraindicated in children below the age of 12 years for the symptomatic treatment of cold.

<u>Children aged 12 years to 18 years:</u>
P. P. Dihydrocodeine is not recommended for use in children aged 12 years to 18 years with compromised respiratory function.

ROUTE OF ADMINISTRATION:
Oral administration

CONTRAINDICATION:

- Hypersensitivity to dihydrocodeine, respiratory depression, obstructive airways disease, in acute alcoholism.
- In children below the age of 12 years for the symptomatic treatment of colds due to an increased risk of developing serious and life- threatening adverse reactions.
- In all paediatric patients (0-18 years of age) who undergo tonsillectomy and/or adenoidectomy for obstructive sleep apnoea syndrome due to increased risk of developing serious and life-threatening adverse reactions.
- In women who are breastfeeding.
- In patients for whom it is known they are CYP2D6 ultra- rapid metabolisers.

WARNING AND PRECAUTIONS:

As dihydrocodeine may bring about histamine release, it should not be given during an asthmatic attack. It should be used with caution in patients who are liable to asthmatic attacks. It should also be given with caution to patients with head injuries and conditions in which intracranial pressure is raised. Dihydrocodeine should be used in reduced dosages in the elderly, in those with hypothyroidism, kidney or liver function impairment.
Prolonged use of dihydrocodeine may produce dependence. However, its addictive potential is very low and it can be safely used more frequently and for longer periods than other more potent narcotic analgesics.

Use in children:

Dihydrocodeine is not recommended for use in children below 4 years old.

CYP2D6 metabolism

Codeine is metabolised by the liver enzyme CYP2D6 into morphine, its active metabolite. If a patient has a deficiency or is completely lacking this enzyme an adequate analgesic effect will not be obtained. Estimates indicate that up to 7% of the Caucasian population may have this deficiency. However, if the patient is an extensive or ultra-rapid metaboliser there is an increased risk of developing side effects of opioid toxicity even at commonly prescribed doses. These patients convert codeine into morphine rapidly resulting in higher than expected serum morphine levels.

General symptoms of opioid toxicity include confusion, somnolence, shallow breathing, small pupils, nausea, vomiting, constipation and lack of appetite. In severe cases this may include symptoms of circulatory and respiratory depression, which may be life-threatening and very rarely fatal. Estimates of prevalence of ultra-rapid metabolisers in different populations are summarised below:

Population	**Prevalence %**
African/Ethiopian	29%
African American	3.4% to 6.5%
Asian	1.2% to 2%
Caucasian	3.6% to 6.5%
Greek	6.0%
Hungarian	1.9%
Northern European	1% to 2%

Post-operative use in children

There have been reports in the published literature that codeine given post-operatively in children after tonsillectomy and/or adenoidectomy for obstructive sleep apnoea, led to rare, but life-threatening adverse events including death. All children received doses of codeine that were within the appropriate dose range; however there was evidence that these children were either ultra-rapid or extensive metabolisers in their ability to metabolise codeine to morphine.

Children with compromised respiratory function

Codeine is not recommended for use in children in whom respiratory function might be compromised including neuromuscular disorders, severe cardiac or respiratory conditions, upper respiratory or lung infections, multiple trauma or extensive surgical procedures. These factors may worsen symptoms of morphine toxicity.

Risks from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of P.P Dihydrocodeine with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate. If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use. If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response. If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response. Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when P.P Dihydrocodeine is used with benzodiazepines. Advice patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Interactions with Other Medicaments).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of P.P Dihydrocodeine with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range. Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue P.P Dihydrocodeine if serotonin syndrome is suspected.

Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience).

INTERACTIONS WITH OTHER MEDICAMENTS:

As with other narcotic analgesics, dihydrocodeine should be used with caution in patients receiving MAOIs. Concurrent treatment with other narcotic analgesics, tranquillizers, sedative-hypnotics, phenothiazines, alcohol and other CNS depressants should be avoided as it may cause an additive CNS depression effect.

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death. The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at µ-receptors, and benzodiazepines interact at $GABA_A$ sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.
Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).
Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue P.P Dihydrocodeine if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

PREGNANCY AND LACTATION:

Pregnancy

Careful consideration should be given before prescribing the product for pregnant patients. Opioid analgesics may depress neonatal respiration and cause withdrawal effects in neonates of dependent mothers. As a precautionary measure, use of P.P Dihydrocodeine should be avoided during the third trimester of pregnancy and during labor.

Breastfeeding

P.P Dihydrocodeine is contraindicated in women during breastfeeding.

At normal therapeutic doses codeine and its active metabolite may be present in breast milk at very low doses and is unlikely to adversely affect the breast fed infant. However, if the patient is an ultra-rapid metaboliser of CYP2D6, higher levels of the active metabolite, morphine, may be present in breast milk and on very rare occasions may result in symptoms of opioid toxicity in the infant, which may be fatal.

ADVERSE EFFECTS:

The common side-effects of dihydrocodeine include constipation, nausea, vomiting, headache and vertigo, especially with higher doses.

Postmarketing Experience:

Serotonin syndrome (See Warnings and Precautions)

Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

OVERDOSAGE AND TREATMENT:

In case of overdosage, gastric lavage should be carried out. Respiratory depression can be treated with naloxone HCl 0.4mg to 2.0mg intravenously, repeated as necessary at 2-3 minutes intervals, up to a total dose of 10mg. Other symptomatic treatment should be instituted as appropriate.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

Dihydrocodeine may cause drowsiness, and, if affected, patients should not drive or operate machinery.

STORAGE CONDITIONS:

Store in a dry place below 30°C

Protect from light.

Keep out of reach of children

PACK SIZES:

Blister of 10 tablets. 3, 6, 9, 10, 25, 50 or 100 blisters in a box.

MANUFACTURER AND PRODUCT REGISTRATION HOLDER:

NORIPHARMA SDN. BHD. 200701034604 (792633-A)

Lot 5030, Jalan Teratai,

5 1/2 Mile Off Jalan Meru,

41050 Klang,

Selangor Darul Ehsan,

Malaysia.

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